10.1.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about sectors.

Unit 10, Lesson 2: Arc Length

Benchmark

MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.

Learning Targets

- ☐ I can define arc length.
- ☐ I can explain how the length of an arc is proportional to the ratio of the central angle to the whole circle, measured in degrees.

10.2.1 Warm-Up: Notice and Wonder

The Chandelier Tree, located in Leggett California, is part of a drive-thru tree park. The tree is believed to have been carved in the early 1930's by Charlie Underwood.

1. What do you notice?

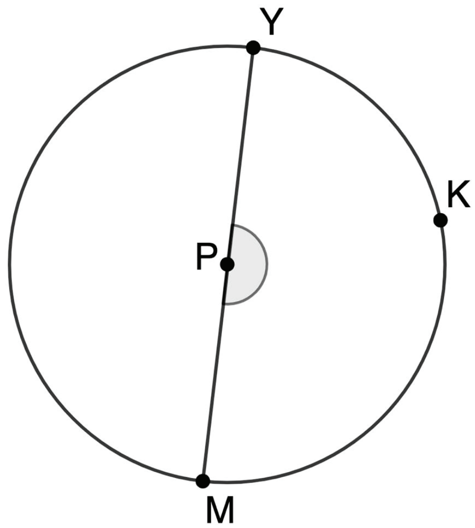
2. What do you wonder?

3. Estimate the diameter of the tree at the base.

4. Explain why you chose that number as your estimate.

10.2.2 Exploration: Arc Length and Proportional Relationships

1. Circle P with point K on the circle and diameter $\overline{YM}$ is shown.



   a. What is $m\angle YPM$?

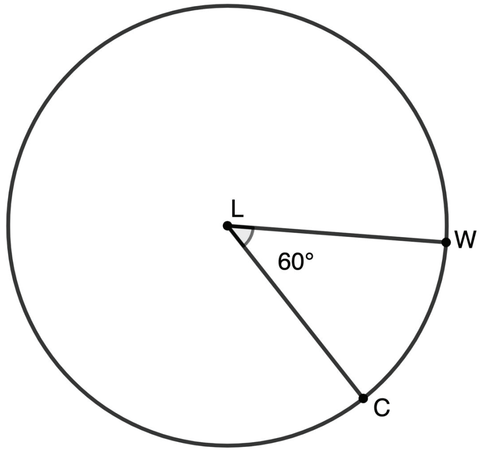
   b. Determine the ratio of the measure of the central angle, $\angle YPM$, to the whole circle. Simplify the ratio.

   c. An arc of a circle is the portion of the circumference between points. Point K is a point on Circle P. Discuss with your partner which part of the circle is arc YKM. Then, trace over this portion of the circle.

   d. What fraction of the circle's circumference does arc YKM represent?

Arc YKM can also be notated as $\overset{\frown}{YKM}$.

2. Circle L is shown where points W and C are on the circle.



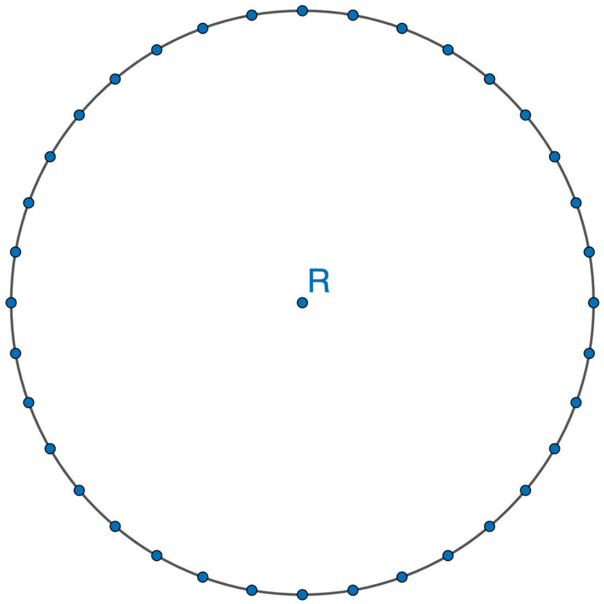
a. What is the ratio of the central angle, $\angle CLW$, to the whole circle? Simplify the ratio.

b. What fraction of the circle's circumference does $\overset{\frown}{WC}$ represent?

3. Verify your answers to problems 1 and 2 with your partner and resolve any differences, if necessary.

4. With your partner, discuss any relationships you observe. Summarize your discussion.

5. On Circle R, the distance between each point measured along the circumference is 1 inch.



a. Mark a point anywhere along the circle's circumference as point T. Then, starting at point T, mark an 8-inch section along the circle's circumference. Label the other endpoint of the 8-inch section as point G.

b. What fraction of the circle's circumference does the length of $\overset{\frown}{TG}$ represent?

c. Draw the central angle $\angle TRG$.

d. What is the measure of the central angle $\angle TRG$?

e. What is the ratio of the central angle to the whole circle?

f. Discuss with your partner the relationship between the length of the arc and the ratio of the central angle to the whole circle.

6. Complete the statements.

The
- ○ arc length
- ○ diameter
- ○ radius

is the distance along the part of a circle's circumference or along any curve. The ratio of the length of an arc to the circumference of a circle is
- ○ the inverse of
- ○ proportional to

the ratio of the central angle to the whole circle.

10.2.3 Activity: Matching Central Angles and Ratios of Arc Length to Circumference

1. Your teacher will give you two sets of cards for this activity. One set shows circles with a central angle. The other set of cards is a set of ratios of the arc length to the circumference.

   a. Work with your partner to match each circle card to the corresponding ratio indicated on the lettered card. Record the matches in the table.

Card Letter	Ratio of the Arc Length to the Circumference	Card Letter	Ratio of the Arc Length to the Circumference
A		F	
B		G	
C		H	
D		I	
E		J	

   b. Summarize how you and your partner were able to match the circles to a corresponding ratio without being given the circumference.

c. Complete the statement.

The ratio of the central angle to the whole circle is the

○ inverse of
○ same as

the ratio of the length of an arc to the circumference.

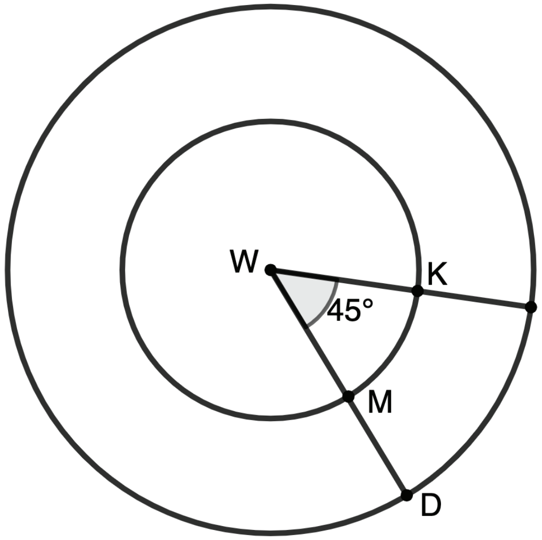
10.2.4 Exploration: Arc Lengths with the Same Central Angle and Different Radii

1. Points D and P lie on Circle W. The radius $\overline{WP}$ measures 5 inches (in.).

   A smaller Circle W with a radius measure of 2 in. rests inside the larger Circle W, where points M and K lie on the smaller circle.

   The smaller Circle W has central angle $\angle KWM$ that measures $45°$ and the larger Circle W has central angle $\angle PWD$ that measures $45°$.

   Use the information to complete the following.



   a. $\widehat{MK}$ and $\widehat{DP}$ are different lengths. Discuss with your partner whether you think the ratio of the arc length and circumference will be the same as or different than the ratio of the central angle of the whole circle.

b. Andrew and Sarah are discussing if the ratio of the arc length to the circumference is the same as the ratio of the central angle to the whole circle for both the smaller and larger circles.

Sarah says, "Yes. They will have the same ratios because they have the same central angle."	Andrew says, "No. They do not have the same ratios because they are different circles with different radii."

Explain who you agree with.

c. Complete the table to support your reasoning in Part B.

	Ratio of Arc Length to Circumference	**Ratio of Central Angle to Whole Circle**
Circle W with radius 2 in.		
Circle W with radius 5 in.		

d. Explain whether your findings in Part C support your answers to Parts A and B. If not, explain why your answers to Parts A and B are incorrect.

10.2.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.